

Analysis of Sequencing Data – Final Assignment

Karolína Šustrová

Methods

Quality Control and Assembly

Raw reads obtained from Nanopore sequencing were processed using Fastplong (version 0.4.1) (Chen, 2023) to remove sequencing adapters and filter out low-quality reads. The filtered reads were assembled into contigs using Flye (version 2.9.2) (Kolmogorov et al., 2019). The assembly process was executed on the Metacentrum computing cluster utilizing a batch job script, with the assembly mode set to high-quality nanopore reads (--nano-hq).

Genome QC and Annotation

The quality and structural integrity of the assembled genome were evaluated using the basic assembly statistic derived from the flye.log output file. Genome completeness was quantitatively assessed using BUSCO (version 5.2.2) (Manni et al., 2021) against the bacteria_odb10 lineage dataset. Finally, the assembled genome was annotated using Prokka (version 1.14.6) (Seemann, 2014) to identify structural genomic features and genes. The specific 16S rRNA were localized by parsing the resulting .gff and .ffn annotation files.

Results

Reads Characteristics

The sequencing reads were evaluated before and after quality control process. The application of Fastplong successfully removed low-quality sequences, which slightly reduced the total number of bases but increased the overall proportion of high-quality bases in the dataset. The detailed characteristics of the raw and processed reads are summarized in Table 1.

Table 1. Characteristics of raw and processed sequencing reads used in this study.

Metric	Raw Reads	Processed reads
Total bases	1.114901 G	1.080357 G
Mean read length	8.599000 K	8.471000 K
Bases ≥ Q10	91.47 %	92.22 %

Genome Assembly Statistics and Annotation

The computational assembly yielded a draft genome consisting of 4 contigs with a total size of 2 758 447 bp and an N50 value of 2 714 464. The BUSCO analysis indicated a high level of genome completeness reaching 97.6 %. Genome statistics of the assembled genome are summarized in Table 2.

Table 2. Genome statistics of the newly assembled genome presented in this study

Feature	Value
Genome size	2 758 447 bp
Number of contigs	4
N50	2 714 464
Mean genome coverage	363
BUSCO completeness	97.6% (C:97.6%, S:96.8%, D:0.8%, F:1.6%, M:0.8%, n:124)

Genome annotation using Prokka identified a total of 2780 annotated genes and 6 rRNA elements. The structural search for 16S rRNA genes revealed two specific localizations on Contig_1: the first at coordinates 3731-5263 (Locus Tag: DPCFIDGP_00004) and the second at coordinates 1195474-1197006 (Locus Tag: DPCFIDGP_01231).

References

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